

Nonlinear Control Of Complex Networks For Minimum Dominating Sets

Abstract

Control theory can be applied to various complex systems such as biological systems. Also, many real systems are nonlinear. Therefore, to control these complex systems we must consider the nonlinear control problem. By construction of interaction matrix of coupled n -dimensional time series in vicinity of each fixed points, we determine driver nodes and their driver functions for controlling the system by the minimum amount of energy required for the control or any cost functions. In addition, with an efficient algorithm such as the minimum dominating set (MDS) approach, we can control complex networks through a small number of the controller. Our results demonstrate nonlinear optimal control theory through MDS controller nodes for different examples and calculate the required energy to control the system. We provide Examples for nonlinear control theory of deterministic and stochastic dynamical systems.

Keywords— complex networks, controllability, minimum dominating sets.

1 Introduction

Control theory can be applied to various complex systems such as biological systems. Also, many real systems are nonlinear therefore, to control these complex systems we must consider the nonlinear control problem. Also with an efficient algorithm such as the minimum dominating set (MDS) approach, we can control the complex networks through a small number of the controller. Our results demonstrate nonlinear control theory through MDS controller nodes for different set examples.

Recently, we can see an increased interest regarding the control of complex dynamic networks as a result of the successful use of the controllability of complex networks in, for example, gene regulatory networks [5], social networks, protein-protein interaction networks [6], neuronal systems [7, 8], food web [9], and power grids [10, 11]. We define a dynamical system that is controllable if we can control the whole system by selecting suitable control input signals which can drive the system from any initial state to any desired final state in finite time [1, 2, 3].

Liu et al [12] developed structural controllability for directed networks and used a maximum matching algorithm to identify the minimum number of unmatched nodes that need to be driven by external signals for the whole system to achieve full control. Unmatched nodes are called driver nodes, and the number of these determines a system's controllability [13].

There are some known techniques which are dealing with the control of linear and nonlinear systems using the explicit dynamical equations with defining a cost (cost-to-go) function. It is known as optimal control theory. In the optimal control theory, we can define a control input signal which satisfies both the initial and final state of each node in the network and minimizes the cost function as well.

Recently it has been shown that the required energy to control a dynamic system exponentially decays with the number of driver nodes, therefore the MDS approach which gives a small number of driver nodes would be an efficient way to control a dynamical system and the control energy in this approach will be relatively small.

Our first main contribution is to retrieve the driver nodes. We assume nodes in an MDS are driver nodes. Also in the MDS model, we consider that every driver node can control all of its outgoing links separately with a specific control input signal. Therefore we can derive this control input $u(x, t)$ for each edge by solving the dynamics.

We also look at the controllability of nonlinear stochastic systems (optimal control), deriving control function $u(x, t)$ for the MDS model.

Optimal control of nonlinear systems in the presence of noise is a very general problem that occurs in many areas of science and engineering. It underlies autonomous system behavior, such as the control of movement and planning of actions of animals and robots, but also, for instance, the optimization of financial investment policies and control of chemical plants.

2 Minimum dominating set control

In network science, our main focus is to utilize a model that can control the whole system with a minimum number of driver nodes. Liu et al. [12] and Nacher and Akutsu [14, 15] addressed this problem for directed networks and undirected networks, respectively. Nacher and Akutsu [14, 15] introduced the MDS model to study the controllability of undirected networks. Also recently, Liu et al [12] showed that the MDS model is capable of controlling an undirected network, and each node in the MDS can control all of its outgoing edges separately.

We consider Graph $G(V, E)$, is an unweighted, undirected graph, where V and E correspond to the set of nodes and the set of edges, respectively. An adjacency matrix A for a graph with n nodes can be used to represent the edges, where $a_{ii} = 0$ (a self-loop is not considered) for $i = 1, 2, \dots, n$, and $a_{ij} = 1$ if there exists an interaction between nodes i and j and $a_{ij} = 0$ otherwise. In a network or a graph, a dominating set is a subset of nodes in a graph $G(V, E)$, such that each node in V is either in that subset or just adjacent to it. Now, such a subset with the minimum cardinality is a minimum dominating set (MDS).

Identifying an MDS is an NP-hard problem [19, 20], however, it can be mapped onto a linear integer programming problem through which even the large scale networks shall be dealt with in relatively short times.

However, to find an MDS we can formalize the problem as an ILP we consider, $x_i = 1$ and $x_i = 0$ indicate that node i is in an MDS and not in an MDS, respectively:

Minimize

$$\sum_{i \in V} x_i \quad (1)$$

Subject to

$$\begin{aligned} x_i + \sum_{(i,j) \in V} a_{ij}x_j &\geq 1 \quad \forall i \in V \\ x_i &\in \{0, 1\} \quad i, j = 1, 2, \dots, n : i \neq j \end{aligned} \quad (2)$$

Where An MDS is given by the set $\{i | x_i = 1\}$.

Examples are illustrated in Fig. 1.

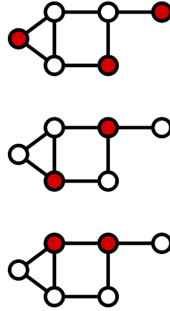


Figure 1: Different configurations of dominating sets in red color; the middle and the below are MDS.

To relate the concepts of MDS and network controllability, a theorem states that supposing that every edge in a network is bi-directional and every node in the MDS can control all of its outgoing links separately, then, the network is structurally controllable by selecting the nodes in the MDS as driver nodes. It can be proved how this framework can guarantee structural controllability. An illustration of the difference between controllability in its basic Kalman sense and in MDS control sense is given in Fig. 2.

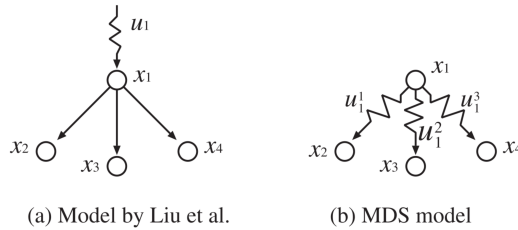


Figure 2: Concepts of maximum matching control (left) and minimum dominating set control (right).

3 Nonlinear controllability

Given the values of the state variable $x(t)$ in any dimensions and the control variable $u(t)$ at time t , the dynamical (state) equation, a differential equation,

$$\dot{x}(t) = f(x(t), u(t), t), \quad x(t_0) = x_0 \quad (3)$$

specifies the instantaneous rate of change in the state variable, where f is a given linear or nonlinear function of x , u , and t and x_0 is the initial value of the state variable. If we know the initial value x_0 and the control trajectory, i.e., the values of $u(t)$ over the whole time interval $t_0 \leq t \leq t_f$, then we can integrate Eq. 3 to get the state trajectory, i.e., the values of $x(t)$ over the same time interval.

We want to choose the control trajectory so that the state and control trajectories maximize (or minimize, depending on the particular problem) the objective functional, or simply the objective function,

$$J = \int_{t_0}^{t_f} R(x(t), u(t), t) dt. \quad (4)$$

In Eq. 4, R is a given function of x , u , and t , which could measure the benefit minus the cost at ending state $x(t_f)$.

3.1 The Optimal Control Problem

Given the preceding definitions, we can state the optimal control problem, which we will be concerned with in this chapter. The problem is to find an admissible control u^* , which minimize the objective function Eq 4. subject to the state equation Eq 3. and the control constraints Eq. 4. We now restate the optimal control problem as:

$$\begin{aligned} \min_{u(t)} \quad & J = \int_{t_0}^{t_f} R(x(t), u(t), t) dt. \\ \text{subject to} \quad & \\ \dot{x}(t) = f(x(t), u(t), t), \quad & x(0) = x_0, \quad x(T) = x_f \end{aligned} \quad (5)$$

The control u^* is called an optimal control and x^* , determined by means of the state equation with $u = u^*$, is called the optimal trajectory or an optimal path. The optimal value $J(u^*)$ of the objective function will be denoted as J^* , and occasionally as $J_{(x_0)}^*$ when we need to emphasize its dependence on the initial state x_0 .

3.2 Coupled dynamical equations for MDS model

For MDS model we implement a non-linear dynamical system to obtain control input signals.

The optimal control is laid out below,
Cost function:

$$J = \frac{1}{2} \left[\int_{t_0}^{t_f} (Tr(u^T(t).u(t)) - x^T(t).x(t) + \frac{1}{2}(x^T(t).x(t))^2) \right]. \quad (6)$$

A system is described by non-linear dynamic:

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + u(t)\mathbf{x}(t) \quad (7)$$

Where the matrix $A \in \mathbf{R}^{n \times n}$ is the adjacency matrix of the n nodes and $\mathbf{x}(t) = [x_1(t), \dots, x_n(t)]^T$ is the $n \times 1$ time-varying state.

The matrix $u(t) \in \mathbf{R}^{n \times n}$ is control input signal matrix. Therefore, $u_{ij} = u_i^j$ is the control input signal from MDS i to a node j which is connected to MDS i with an edge. Other elements in matrix u is 0.

The optimal control input, minimizes the cost function and satisfies an arbitrary initial condition and an arbitrary final condition if the system is controllable.

There are similarities between our approach and directly manipulating the elements of the adjacency matrix. In the second approach the topology in terms of connectivity at any point in time t and the weight of each link can vary. So for a network with N nodes, the adjacency matrix is,

$$A_{N,N}(t) = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1N}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2,N}(t) \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1}(t) & a_{N2}(t) & \cdots & a_{NN}(t) \end{pmatrix} \quad (8)$$

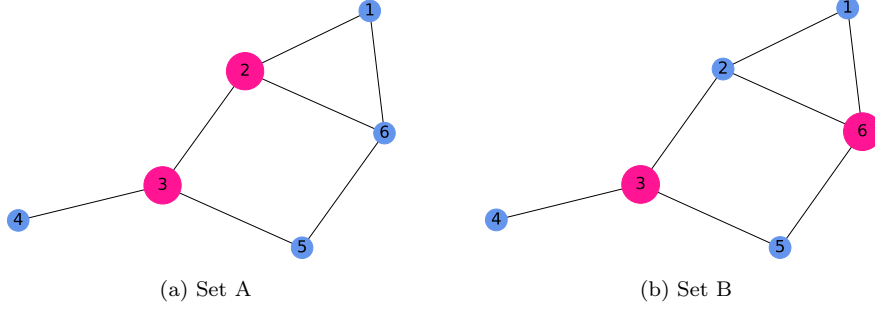


Figure 3: Two different minimum dominating sets, the driver nodes are highlighted in red.

where $a_{ij}(t)$ represents the edge connectivity and weight between node i and j . In order to represent real edge weights as well as connectivity, we let $a_{ij}(t)$ be a real number between 0 and 1 (i.e., $a_{ij}(t) \in (0, 1]$ if link from node i to j exists, otherwise $a_{ij}(t) = 0$).

3.3 Numerical Results

In this section, we demonstrate the control input signal u^* for two examples in the following.

We illustrate our approach to control a complex dynamical network towards the desired state. For the first example we consider one of the nodes in the system is in an undesired states and our purpose is to bring this node back to the our desired state by exercising control inputs (i.e. in the first example node 5 is in the undesired state). Our approach relies on solving differential equations of any given dynamics in the vicinity of a desired final state. We used GEKKO Optimization Suite [16]. GEKKO is a Python package for machine learning and optimization of mixed-integer and differential-algebraic equations. It is coupled with large-scale solvers for linear, quadratic, nonlinear, and mix integer programming (LP, QP, NLP, MILP, MINLP).

In addition, we utilized MDS algorithm that allows one to determine – for the given dynamics and topology – a small set of the required driver nodes.

3.3.1 Example 1:

First, we consider a network with 6 nodes. The networks are illustrated in Fig. 3. Our purpose is to calculate the energy for both sets A and B in Fig. 3.

Energy is,

$$E = \sum_{i=1}^n |u_i|^2 \quad (9)$$

First, from MDS model we retrieved the minimum dominating set for set A. We have two minimum dominating set for this example, $\{2, 3\}$ and $\{3, 6\}$.

The corresponding MDS model equations for network in Set A are:

$$\begin{aligned} \frac{dx_1(t)}{dt} &= x_2(t) + x_6(t) + u_2^1(t)x_1(t) \\ \frac{dx_2(t)}{dt} &= x_1(t) + x_3(t) + x_6(t) + u_3^2(t)x_2(t) \\ \frac{dx_3(t)}{dt} &= x_2(t) + x_4(t) + x_5(t) + u_3^2(t)x_3(t) \\ \frac{dx_4(t)}{dt} &= x_3(t) + u_3^4(t)x_4(t) \\ \frac{dx_5(t)}{dt} &= x_3(t) + x_6(t) + u_3^5(t)x_5(t) \\ \frac{dx_6(t)}{dt} &= x_1(t) + x_2(t) + x_5(t) + u_2^6(t)x_6(t) \end{aligned} \quad (10)$$

where x_i is the time-varying state of the i th node, and $u_{i'}^j$ is the control input from i' th MDS to the j th node. For solving the differential equations in Eq 6. we used GEKKO Optimization Suite [16].

By integrating Eq. (10) for a network in Fig. 3 and initial condition, we derive the temporal evolution of variables x_i and $u_{i'}^j$ which fully determine the system's dynamics.

The corresponding MDS model equations for network in Set B are:

$$\begin{aligned}
\frac{dx_1(t)}{dt} &= x_2(t) + x_6(t) + u_6^1(t)x_1(t) \\
\frac{dx_2(t)}{dt} &= x_1(t) + x_3(t) + x_6(t) + (u_3^2(t) + u_6^2(t))x_2(t) \\
\frac{dx_3(t)}{dt} &= x_2(t) + x_4(t) + x_5(t) \\
\frac{dx_4(t)}{dt} &= x_3(t) + u_3^4(t)x_4(t) \\
\frac{dx_5(t)}{dt} &= x_3(t) + x_6(t) + (u_3^5(t) + u_6^5(t))x_5(t) \\
\frac{dx_6(t)}{dt} &= x_1(t) + x_2(t) + x_5(t)
\end{aligned} \tag{11}$$

The time-varying state and optimal control input for both set A and B are illustrated in Figs. 4 and 5.

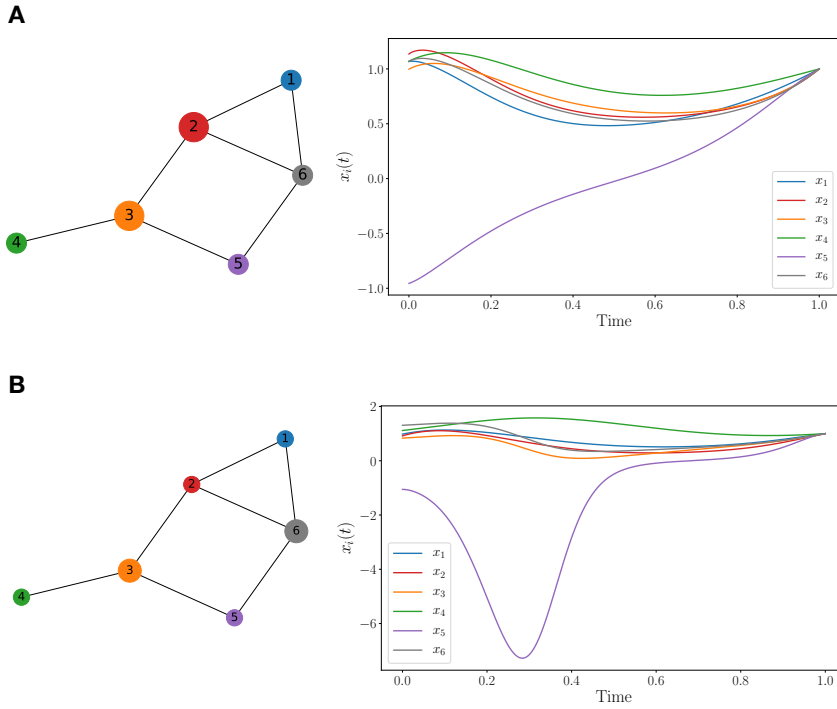


Figure 4: **(A)** Example network consisting of the 6 nodes, where nodes 2 and 3 are the drivers in Set A. The state evolution is shown where the initial condition is the $1 + \mathcal{N}(0, 0.01)$ and the final state for each target node is $x_i = 1$. **(B)** Example network consisting of the 6 nodes, where nodes 3 and 6 are the drivers in Set B. The state evolution is shown where the initial condition is the $1 + \mathcal{N}(1, 0)$ and the final state for each target node is $x_i = 1$.

The square of the magnitude of the control input is equal to energy as it was mentioned in Eq. 9. The energy, or the control effort, is found by integrating the square of the magnitude of the control input. For the first case $E \approx 35$ (a.u) and for the second case is $E \approx 63$ (a.u). Although we should consider that because the initial condition is a random variable the fluctuation of energy is significant. The temporal evolution of energy for both set A and set B are illustrated in Fig. (6).

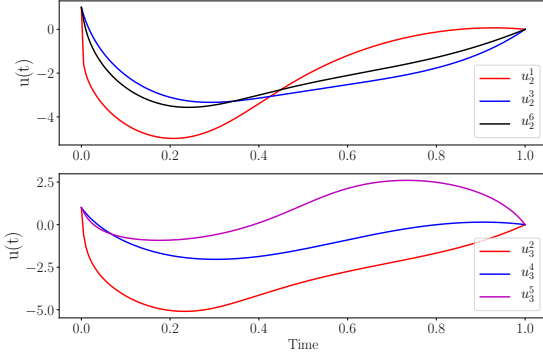
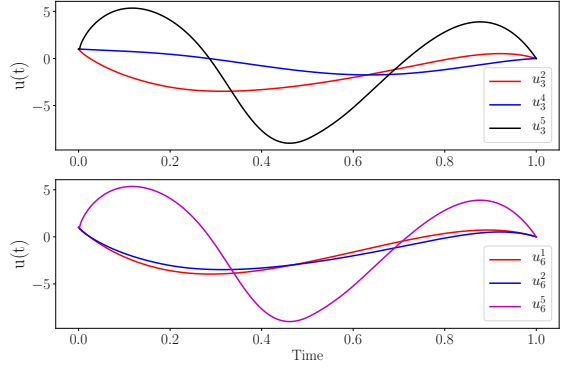
A**B**

Figure 5: **(A)** The optimal control input for set A. **(B)** The optimal control input for set B.

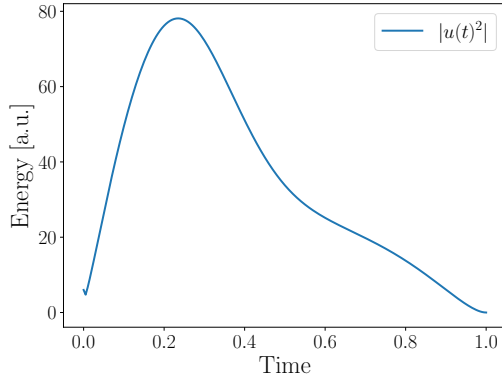
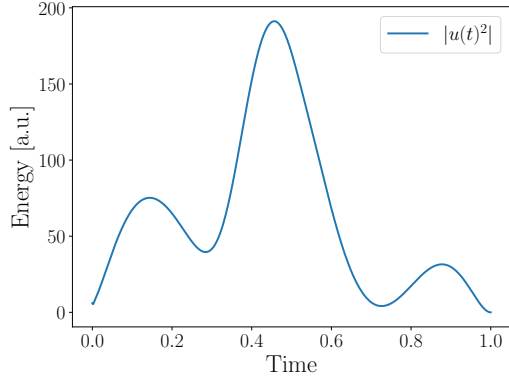
A**B**

Figure 6: **(A)** The square of the magnitude of the control input. The energy, or the control effort, is found by integrating the square of the magnitude of the control input for set A. For this case, $E \int |u(t)| \approx 35(a.u.)$. **(B)** The square of the magnitude of the control input for set B. For this case, $E \int |u(t)| \approx 63(a.u.)$.

3.3.2 Example 2:

In this section we obtain optimal control input and energy for another network consisting of the 10 nodes. For this example we considered all of the nodes except nodes in MDS are in the undesired states and our purpose is to bring all of the node back to the desired states. The network is illustrated in Fig. 7

The corresponding MDS model equations for network in Set A are:

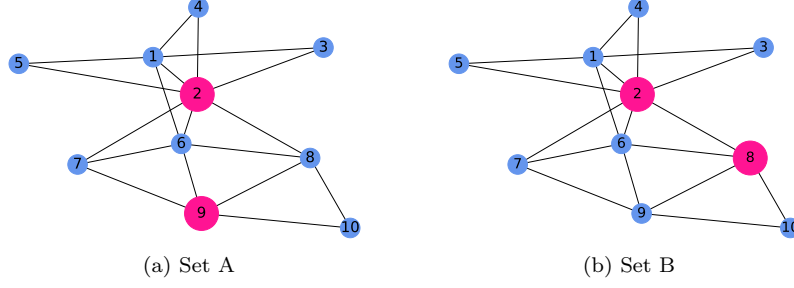


Figure 7: Two different minimum dominating sets, the driver nodes are highlighted in red.

$$\begin{aligned}
\frac{dx_1(t)}{dt} &= x_2(t) + x_3(t) + x_4(t) + x_5(t) + x_6(t) + u_2^1(t)x_1(t) \\
\frac{dx_2(t)}{dt} &= x_1(t) + x_3(t) + x_4(t) + x_5(t) + x_6(t) + x_7(t) + x_8(t) \\
\frac{dx_3(t)}{dt} &= x_1(t) + x_2(t) + (u_2^3(t))x_3(t) \\
\frac{dx_4(t)}{dt} &= x_1(t) + x_2(t) + u_2^4(t)x_4(t) \\
\frac{dx_5(t)}{dt} &= x_1(t) + x_2(t) + u_2^5(t)x_5(t) \\
\frac{dx_6(t)}{dt} &= x_1(t) + x_2(t) + x_7(t) + x_8(t) + x_9(t) + (u_2^6(t) + u_9^6(t))x_6(t) \\
\frac{dx_7(t)}{dt} &= x_2(t) + x_6(t) + x_9(t) + (u_2^7(t) + u_9^7(t))x_7(t) \\
\frac{dx_8(t)}{dt} &= x_2(t) + x_9(t) + x_6(t) + x_{10}(t) + (u_9^8(t) + u_2^8(t))x_8(t) \\
\frac{dx_9(t)}{dt} &= x_6(t) + x_7(t) + x_8(t) + x_{10}(t) \\
\frac{dx_{10}(t)}{dt} &= x_8(t) + x_9(t) + u_9^{10}(t)x_{10}(t)
\end{aligned} \tag{12}$$

By integrating Eq. (12) for a network in Fig. 7 and the initial condition, we derive the temporal evolution of variables x_i and u_i^j which fully determine the system's dynamics.

The corresponding MDS model equations for network in Set B are:

$$\begin{aligned}
\frac{dx_1(t)}{dt} &= x_2(t) + x_3(t) + x_4(t) + x_5(t) + x_6(t) + u_2^1(t)x_1(t) \\
\frac{dx_2(t)}{dt} &= x_1(t) + x_3(t) + x_4(t) + x_5(t) + x_6(t) + x_7(t) + x_8(t) + u_8^2(t)x_2(t) \\
\frac{dx_3(t)}{dt} &= x_1(t) + x_2(t) + u_2^3(t)x_3(t) \\
\frac{dx_4(t)}{dt} &= x_1(t) + x_2(t) + u_2^4(t)x_4(t) \\
\frac{dx_5(t)}{dt} &= x_1(t) + x_2(t) + u_2^5(t)x_5(t) \\
\frac{dx_6(t)}{dt} &= x_1(t) + x_2(t) + x_7(t) + x_8(t) + x_9(t) + (u_2^6(t) + u_8^6(t))x_6(t) \\
\frac{dx_7(t)}{dt} &= x_2(t) + x_6(t) + x_9(t) + u_2^7(t)x_7(t) \\
\frac{dx_8(t)}{dt} &= x_2(t) + x_9(t) + x_6(t) + x_{10}(t) + u_2^8(t)x_8(t) \\
\frac{dx_9(t)}{dt} &= x_6(t) + x_7(t) + x_8(t) + x_{10}(t) \\
\frac{dx_{10}(t)}{dt} &= x_8(t) + x_9(t) + u_8^{10}(t)x_{10}(t)
\end{aligned} \tag{13}$$

The time-varying state and optimal control input for both sets are illustrated in Figs. 8 and 9.

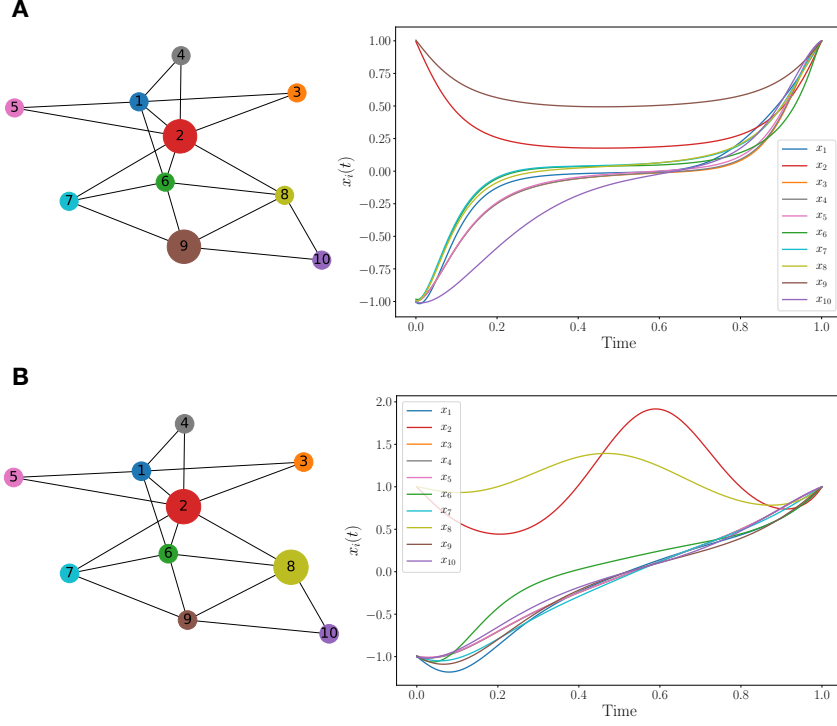


Figure 8: **(A)** Example network consisting of the 10 nodes, where nodes 2 and 9 are the drivers in Set A. The state evolution is shown where the initial condition is the $1 + \mathcal{N}(0, 0.01)$ and the final state for each target node is $x_i = 1$. **(B)** Example network consisting of the 10 nodes, where nodes 2 and 8 are the drivers in Set B. The state evolution is shown where the initial condition is the $1 + \mathcal{N}(0, 0.01)$ and the final state for each target node is $x_i = 1$.

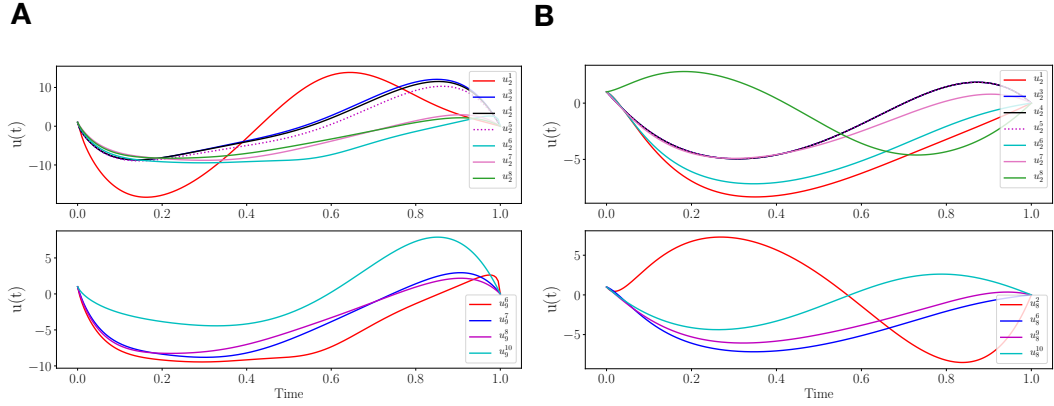


Figure 9: **(A)** The optimal control input for set A. **(B)** The optimal control input for set B.

The energy, or the control effort, is found by integrating the square of the magnitude of the control input. For the first case $E \approx 501$ (a.u) and for the second case is $E \approx 180$ (a.u). Although we should consider that because the initial condition is a random variable the fluctuation of energy is significant. The temporal evolution of energy for both set A and set B are illustrated in Fig. (10).

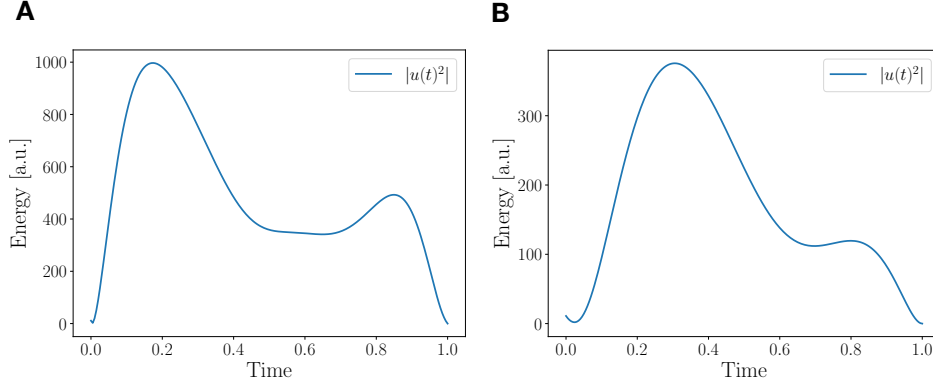


Figure 10: **(A)** The square of the magnitude of the control input for set A. The energy, or the control effort, is found by integrating the square of the magnitude of the control input. For this case, $E \int |u(t)| \approx 501(a.u.)$. **(B)** The square of the magnitude of the control input for set B. For this case, $E \int |u(t)| \approx 180(a.u.)$.

4 Optimal control of nonlinear stochastic systems

Optimal control of nonlinear systems in the presence of noise is a very general problem that occurs in many areas of science and engineering.

The problem is simply stated: given that the system is in this configuration at this time, what is the optimal course of action to reach a goal state at some future time?

Recently, it has been found that a class of continuous non-linear stochastic control problems can be solved more efficiently than the general case (PRL95, 200201 (2005)). These are control problems with a finite time horizon, where the control acts linearly and additive on the dynamics and the cost of the control is quadratic. These control problems can have both time-dependent and time-independent solutions.

Here we consider the simplest stochastic dynamical equation with given additive control and noise terms. Let x be an n -dimensional stochastic variable that is subject to the stochastic differential equation,

$$d\mathbf{x}(t) = A\mathbf{x}(t)dt + u(\mathbf{x}, t)\mathbf{x}(t)dt + \sqrt{dt}\xi \quad (14)$$

where,

$$\xi_i = \mathcal{N}(0, 1) \quad (15)$$

Define cost-to-go function as,

$$C = \left\langle \frac{1}{2} \int_{t_0}^{t_f} (Tr(u^T(t).u(t)) + x^T(t).x(t) + \frac{1}{2}(x^T(t).x(t))^2) \right\rangle_x. \quad (16)$$

The notation $u(t)$ denotes an input signal on the time t .

The objective is to find the controller $u = u^*(x, t)$ for the system in Eq. 13 such that the cost function $C(x(t), u(t), t)$ is minimized.

The value function $J(x, t)$ will be,

$$J(x, t) = \min_{u(t)} C(x(t), u(t), t) \quad (17)$$

4.1 Numerical Examples for Stochastic case

In this section, some cases of stochastic optimal control problems are investigated. Furthermore, to highlight the applications of stochastic optimal control problems we consider the differential equations of Eqs. 10 and Eqs. 11 with the additive noise which was discussed in Eq. 13.

The graphs of state evolution $x(t)$ and optimal control variable $u(x, t)$ for set A in Fig. 3 are given in Fig. 11 and Fig. 12.

Also, The graphs of state evolution $x(t)$ and optimal control variable $u(x, t)$ for set B in Fig. 3 are given in Fig. 13 and Fig. 14.

As we can see in Fig. 13 and Fig. 15 we can conclude that the dynamical system in set B takes a longer time to be controlled by optimal control input u^* from nodes in MDS. Also, the stochasticity of $u(x, t)$ as a result of stochasticity in x is obviously demonstrated as a result of stochasticity in x .

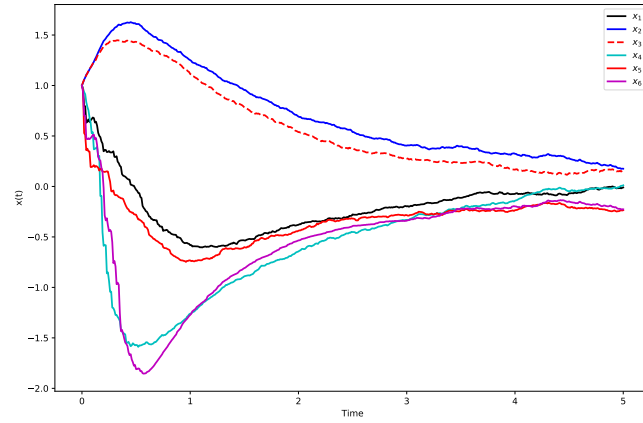


Figure 11: The state evolution is shown where the initial condition is the $1 + \mathcal{N}(0, 0.01)$ and the final state for each target node is $x_i = 0$.

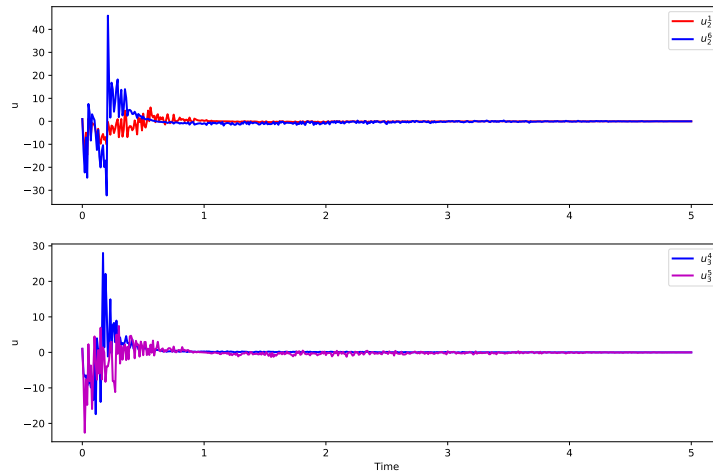


Figure 12: The stochastic optimal control input.

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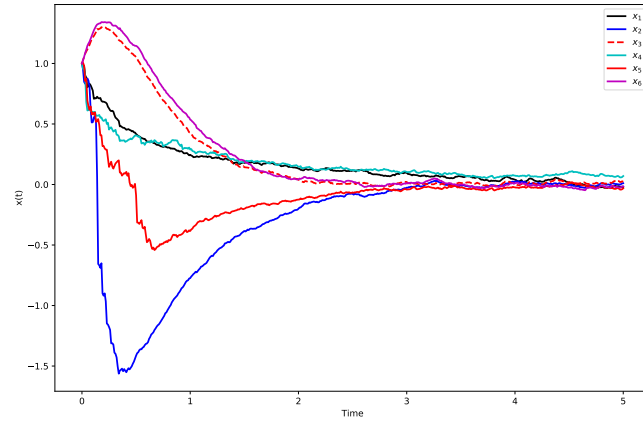


Figure 13: The state evolution is shown where the initial condition is the $1 + \mathcal{N}(0, 0.01)$ and the final state for each target node is $x_i = 0$.

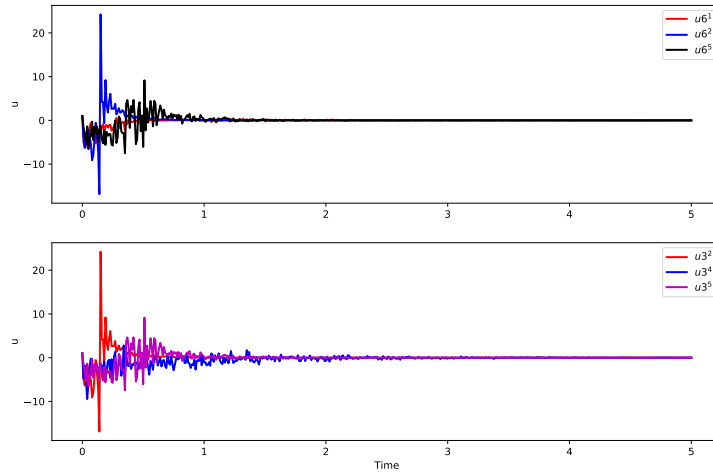


Figure 14: The stochastic optimal control input.

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